

# Gold Future/ETF Analysis

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2025

| Maturity | Slope   | Intercept                | $R^2$   | RMSE    |
|----------|---------|--------------------------|---------|---------|
| 1-Month  | 0.99962 | $1.2811 \times 10^{-7}$  | 0.94937 | 0.00221 |
| 2-Month  | 1.00241 | $-1.3769 \times 10^{-6}$ | 0.94710 | 0.00227 |
| 6-Month  | 1.00322 | $8.1922 \times 10^{-7}$  | 0.94068 | 0.00242 |
| 12-Month | 0.94779 | $2.2525 \times 10^{-5}$  | 0.83986 | 0.00409 |

Table 1: New Data: A summary of the regression coefficients and measures of goodness of fit for regressing one-day returns of 1, 2, 6 and 12-month futures on 1-day returns of spot gold.

The slopes for the 1, 2, and 6-month contracts are all very close to 1, indicating a strong correlation with spot gold returns. The 12-month contract, however, has a noticeably lower slope of 0.94779 and a significantly lower  $R^2$  of 0.83986. This may be a result of the lower liquidity and trading volume of the longer-dated contract, making it less sensitive to daily spot price movements. The great difference in all statistics, though, could also indicate that the data for the 12-month contract data could be corrupted, recorded incorrectly, or otherwise flawed.

| Maturity | Slope   | Intercept                | $R^2$   | RMSE    |
|----------|---------|--------------------------|---------|---------|
| 1-Month  | 0.94314 | $2.4191 \times 10^{-5}$  | 0.80947 | 0.00530 |
| 2-Month  | 0.94301 | $1.0000 \times 10^{-5}$  | 0.80911 | 0.00530 |
| 6-Month  | 0.94348 | $3.1797 \times 10^{-5}$  | 0.80934 | 0.00530 |
| 12-Month | 0.94358 | $-5.2333 \times 10^{-5}$ | 0.80984 | 0.00529 |

Table 2: Leung & Ward (2015) Data: Regression coefficients for one-day futures returns on one-day spot gold returns.

The slopes are also very close to 1, but consistently lower than the newer data. The  $R^2$  values are also significantly lower than the newer data. This may be due to the smaller sample size of the older data and lower liquidity. However, the much greater consistency of the Leung & Ward slopes and the

lower  $R^2$  values could also indicate a procedural difference in how the statistics were calculated.

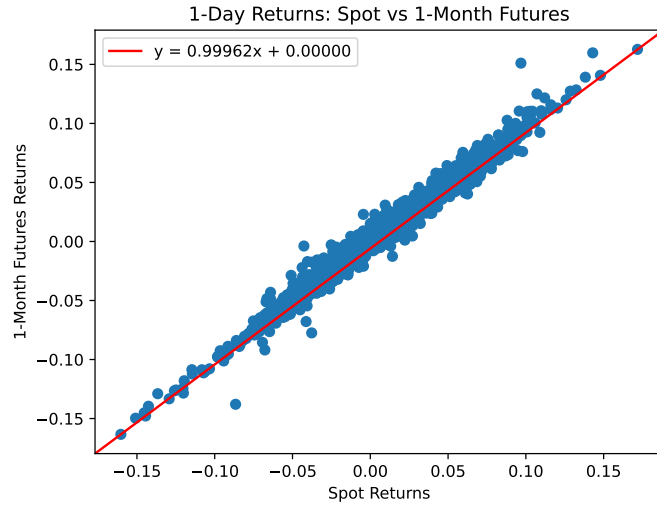


Figure 1: Scatter plot of one-day returns for spot gold versus 1-month gold futures.

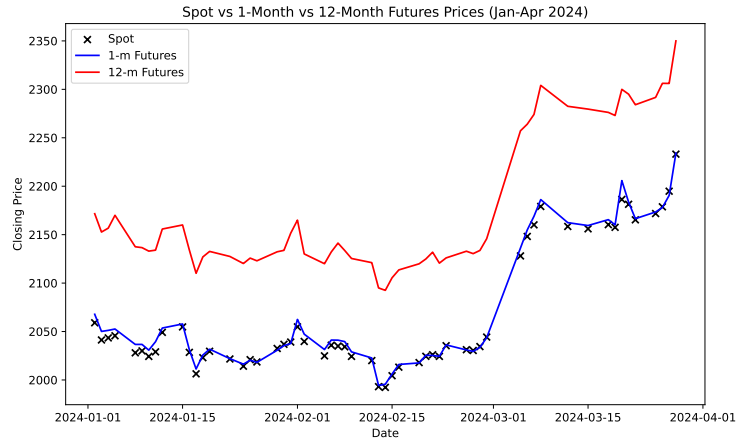


Figure 2: Comparison of closing prices for spot gold, 1-month futures, and 12-month futures in early 2024. The 1-month contract tracks the spot price closely, while the 12-month contract maintains a significant premium.

|              | Days | 1-Month | 2-Month | 6-Month | 12-Month |
|--------------|------|---------|---------|---------|----------|
| <b>Slope</b> | 1    | 0.99962 | 1.00241 | 1.00322 | 0.94779  |
|              | 5    | 1.00741 | 1.01021 | 1.01058 | 0.98753  |
|              | 10   | 1.00224 | 1.00399 | 1.00337 | 0.98967  |
|              | 15   | 1.00312 | 1.00494 | 1.00331 | 0.99205  |
| $R^2$        | 1    | 0.94937 | 0.94710 | 0.94068 | 0.83986  |
|              | 5    | 0.98419 | 0.98398 | 0.98117 | 0.95670  |
|              | 10   | 0.99152 | 0.99101 | 0.98776 | 0.97473  |
|              | 15   | 0.99386 | 0.99330 | 0.98994 | 0.98068  |

Table 3: New Data: Slopes and  $R^2$  from the regressions of futures returns versus gold returns over different holding periods.

The slopes and  $R^2$  values generally increase with longer holding periods but decrease with longer-dated contracts. This suggests that futures returns are correlated with spot returns over longer horizons, with shorter measurement periods capturing more noise.

|              | Days | 1-Month | 2-Month | 6-Month | 12-Month |
|--------------|------|---------|---------|---------|----------|
| <b>Slope</b> | 1    | 0.94314 | 0.94301 | 0.94348 | 0.94358  |
|              | 5    | 0.99805 | 0.99752 | 0.99757 | 0.99715  |
|              | 10   | 1.01075 | 1.01020 | 1.00970 | 1.00841  |
|              | 15   | 0.99448 | 0.99461 | 0.99431 | 0.99345  |
| $R^2$        | 1    | 0.80947 | 0.80911 | 0.80934 | 0.80984  |
|              | 5    | 0.97154 | 0.97158 | 0.97194 | 0.97165  |
|              | 10   | 0.98516 | 0.98530 | 0.98572 | 0.98550  |
|              | 15   | 0.98704 | 0.98691 | 0.98725 | 0.98713  |

Table 4: Leung & Ward (2015) Data: Slopes and  $R^2$  from regressions of futures returns vs. gold returns over different holding periods.

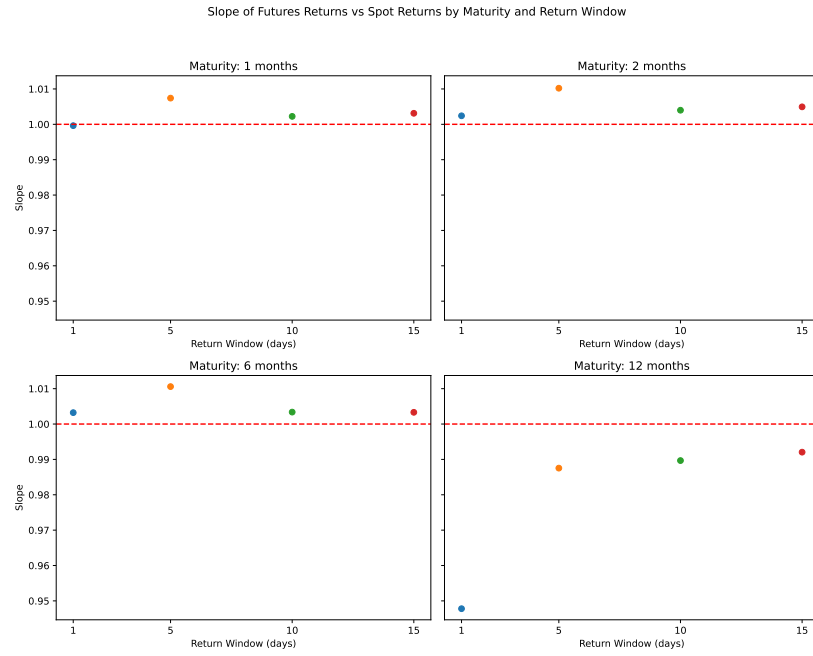


Figure 3: Regression slope of futures vs. spot returns, analyzed by contract maturity and the return window length in days. For most maturities, the slope approaches 1 as the return window lengthens, but the 12-month contract is still significantly different from 1 compared to the other maturities.

The out of sample period is defined as starting on January 1, 2024 and is approximately 10% of the data. The RMSEs are calculated in dollar terms assuming a notional investment of \$1000 in the spot gold price. The weights are calculated using a constrained least squares regression that forces the weights to sum to one.

| <b>Portfolio</b>    | $w_0$   | $w_1$   | $w_2$    | $w_6$    | $w_{12}$ | <b>RMSE<sub>in</sub></b> | <b>RMSE<sub>out</sub></b> |
|---------------------|---------|---------|----------|----------|----------|--------------------------|---------------------------|
| 1-m                 | 0.00258 | 0.99741 | —        | —        | —        | \$2.67                   | \$5.12                    |
| 2-m                 | 0.00679 | —       | 0.99322  | —        | —        | \$3.70                   | \$5.69                    |
| 6-m                 | 0.02215 | —       | —        | 0.97785  | —        | \$10.07                  | \$13.34                   |
| 12-m                | 0.04517 | —       | —        | —        | 0.95483  | \$18.80                  | \$17.72                   |
| 1-m, 2-m            | 0.00170 | 1.21350 | -0.21521 | —        | —        | \$2.62                   | \$5.03                    |
| 1-m, 6-m            | 0.00168 | 1.04713 | —        | -0.04881 | —        | \$2.63                   | \$4.86                    |
| 1-m, 12-m           | 0.00192 | 1.01384 | —        | —        | -0.01576 | \$2.68                   | \$4.91                    |
| 2-m, 6-m            | 0.00333 | —       | 1.23020  | -0.23352 | —        | \$3.21                   | \$4.62                    |
| 2-m, 12-m           | 0.00311 | —       | 1.09427  | —        | -0.09738 | \$3.29                   | \$4.94                    |
| 6-m, 12-m           | 0.00444 | —       | —        | 1.76141  | -0.76585 | \$6.08                   | \$11.87                   |
| 1-m, 2-m, 6-m       | 0.00159 | 1.14787 | -0.12240 | -0.02706 | —        | \$2.62                   | \$4.93                    |
| 1-m, 2-m, 12-m      | 0.00173 | 1.24621 | -0.25302 | —        | 0.00508  | \$2.66                   | \$4.96                    |
| 1-m, 6-m, 12-m      | 0.00201 | 1.10693 | —        | -0.17570 | 0.06676  | \$2.64                   | \$4.52                    |
| 2-m, 6-m, 12-m      | 0.00329 | —       | 1.22206  | -0.22099 | -0.00436 | \$3.24                   | \$4.43                    |
| 1-m, 2-m, 6-m, 12-m | 0.00190 | 1.22259 | -0.14302 | -0.14612 | 0.06465  | \$2.63                   | \$4.60                    |

Table 5: New Data: Weights and in/out of sample RMSEs for portfolios of 1, 2, 3, and 4 futures contracts.

The results show that using a single futures contract provides a reasonable replication, but performance diminishes as the maturity of the contract increases (out-of-sample RMSE rises from \$5.12 to \$17.72). Adding a second futures contract generally improves the out-of-sample RMSE, with the 2-month and 6-month combination performing best at \$4.62. The best-performing portfolio overall is the three-contract combination of 2-month, 6-month, and 12-month futures, which achieved an out-of-sample RMSE of \$4.43. Notably, several of these futures portfolios outperform the GLD ETF, which has an out-of-sample RMSE of \$6.00 over the same period.

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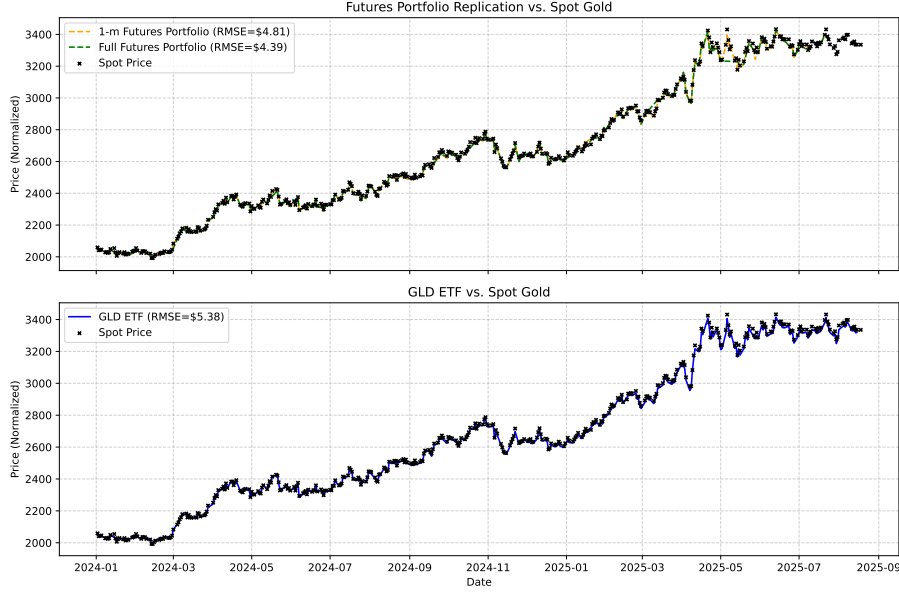


Figure 4: Visual comparison of tracking performance in early 2024. The top panel shows replication of spot gold using futures portfolios, while the bottom panel shows the GLD ETF’s tracking of spot gold. The futures portfolios achieve a lower RMSE in this period than GLD.

| Portfolio | $w_0$    | $w_{short}$ | $w_{long}$ | RMSE <sub>in</sub> | RMSE <sub>out</sub> |
|-----------|----------|-------------|------------|--------------------|---------------------|
| 1-m       | -0.01071 | 1.01071     | —          | \$6.63             | \$3.34              |
| 2-m       | -0.04835 | 1.04835     | —          | \$12.22            | \$6.12              |
| 12-m      | -0.06842 | 1.06842     | —          | \$15.11            | \$5.71              |
| 1-m, 2-m  | -0.00088 | 1.27315     | -0.27227   | \$6.27             | \$2.79              |
| 1-m, 6-m  | -0.00171 | 1.23899     | -0.23729   | \$6.27             | \$2.98              |
| 1-m, 12-m | -0.00021 | 1.19336     | -0.19315   | \$6.29             | \$3.00              |
| 2-m, 6-m  | -0.04079 | 5.06602     | -4.02523   | \$10.27            | \$9.37              |

Table 6: Leung & Ward (2015) Data: Weights and in/out of sample RMSEs for portfolios of 1 and 2 futures contracts.

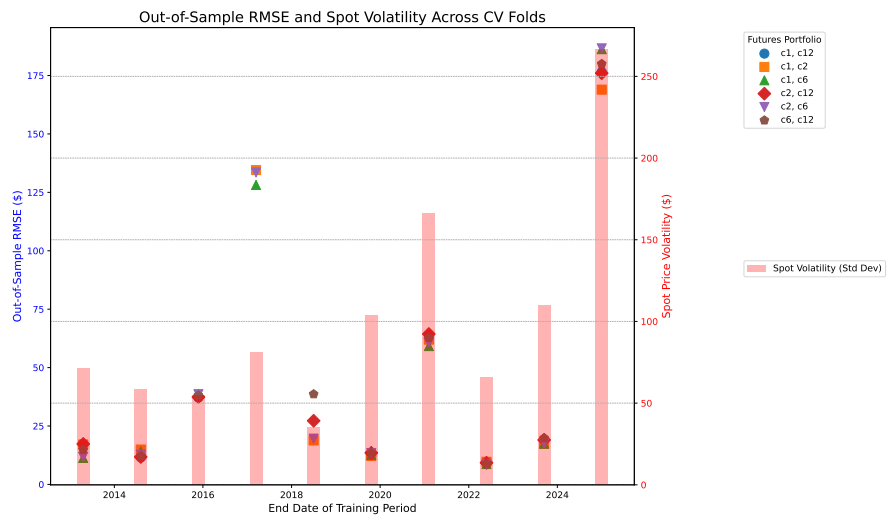


Figure 5: Out-of-sample RMSE for various futures portfolios across different cross-validation training periods. The pink bars represent the spot price volatility during each out-of-sample period.

| (L)ETF | Leverage ( $\beta$ ) | Slope    | Intercept              | t-stat | p-value | $R^2$  | RMSE    |
|--------|----------------------|----------|------------------------|--------|---------|--------|---------|
| GLD    | 1                    | 1.00431  | $-1.30 \times 10^{-5}$ | 2.039  | 0.042   | 0.9780 | 0.00164 |
| UGL    | 2                    | 2.01860  | $-2.10 \times 10^{-4}$ | 2.615  | 0.009   | 0.9688 | 0.00326 |
| GLL    | -2                   | -2.01174 | $1.96 \times 10^{-4}$  | -1.663 | 0.097   | 0.9690 | 0.00323 |
| DGLD   | -2                   | -1.97424 | $9.85 \times 10^{-5}$  | 0.499  | 0.618   | 0.4156 | 0.02058 |
| UGLDF  | 3                    | 3.00291  | $4.25 \times 10^{-4}$  | 0.026  | 0.979   | 0.2501 | 0.04519 |

Table 7: New Data: Regression coefficients for one-day returns of (L)ETFs versus spot gold.

The regression results indicate that GLD, UGL, and GLL are effective at tracking their underlying benchmarks, with high  $R^2$  values and slopes very close to their target betas. However, the low p-values for GLD (0.042) and UGL (0.009) suggest that their slopes are statistically different from the target betas of 1 and -1, respectively. DGLD and UGLDF perform poorly, with very low  $R^2$  values (0.4156 and 0.2501), indicating they do not reliably track their leveraged benchmarks.

| (L)ETF | Leverage ( $\beta$ ) | Slope    | Intercept                | t-stat   | p-value | $R^2$   | RMSE    |
|--------|----------------------|----------|--------------------------|----------|---------|---------|---------|
| GLD    | 1                    | 1.00540  | $-1.6427 \times 10^{-6}$ | 1.42692  | 0.15382 | 0.98060 | 0.00163 |
| UGL    | 2                    | 2.00572  | $-1.3107 \times 10^{-4}$ | 0.73319  | 0.46351 | 0.96257 | 0.00350 |
| GLL    | -2                   | -2.00609 | $1.2541 \times 10^{-4}$  | -0.76019 | 0.44723 | 0.96317 | 0.00346 |
| UGLDF  | 3                    | 3.01189  | $-1.9048 \times 10^{-4}$ | 0.11181  | 0.91099 | 0.28014 | 0.04369 |

Table 8: Leung & Ward (2015) Data: Regression coefficients for one-day returns of (L)ETFs versus spot gold.

## 1 Dynamic Replication

Table 9 displays the tracking error (RMSE) of the actual LETFs against the theoretical leveraged benchmark. This establishes the baseline tracking error inherent in the ETF product itself before comparing it to our replication strategies.

| Table 9: LETF vs. Leveraged Benchmark RMSE |      |                          |
|--------------------------------------------|------|--------------------------|
| ETF                                        | Beta | RMSE (LETF vs Benchmark) |
| GLD                                        | 1    | \$54.78                  |
| UGL                                        | 2    | \$1119.09                |
| GLL                                        | -2   | \$101.39                 |
| UGLDF                                      | 3    | \$1375.81                |



Table 10 summarizes the Root Mean Squared Error (RMSE) for our dynamic replication strategies across different futures maturities.

Table 10: Dynamic Replication RMSE by Futures Maturity (Replicated Portfolio vs Benchmark)

| <b>ETF Target</b> | <b>Beta</b> | <b>1-Month</b> | <b>2-Month</b> | <b>6-Month</b> | <b>12-Month</b> |
|-------------------|-------------|----------------|----------------|----------------|-----------------|
| GLD               | 1           | \$113.96       | \$118.20       | \$119.39       | \$288.40        |
| UGL               | 2           | \$439.63       | \$425.96       | \$395.99       | \$567.73        |
| GLL               | -2          | \$121.88       | \$123.16       | \$127.15       | \$105.97        |
| DGLD              | -3          | \$128.28       | \$129.43       | \$133.21       | \$100.79        |
| UGLDF             | 3           | \$720.58       | \$740.78       | \$718.59       | \$529.40        |

## 2 Dynamic Replication Plots

The following figures illustrate the dynamic replication performance across different futures maturities (1, 2, 6, and 12 months). Each figure displays a 2x2 grid for a specific ETF. It should be noted that the time periods shown in the plots differ between each beta, so the RMSE value are not directly comparable across the different ETFs.

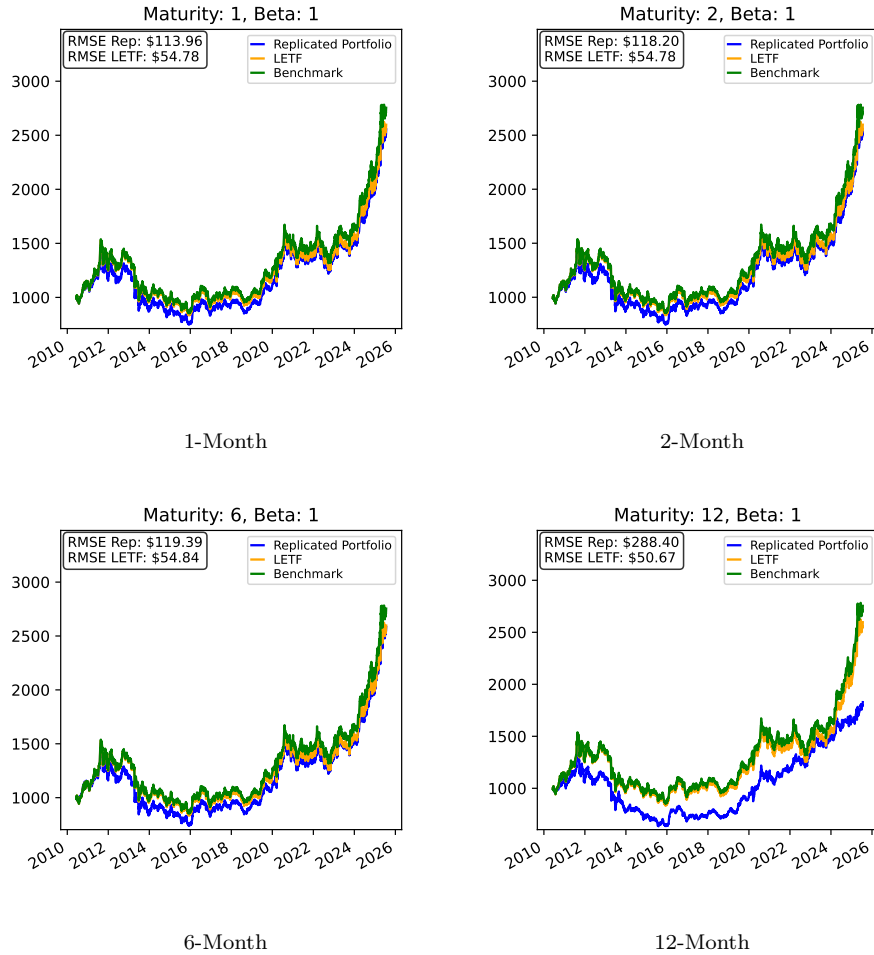


Figure 6: **GLD (1x) Replication**: The GLD ETF performs better than the replicated portfolios, but the shorter-term maturities (1m, 2m, and 6m) show reasonable performance.

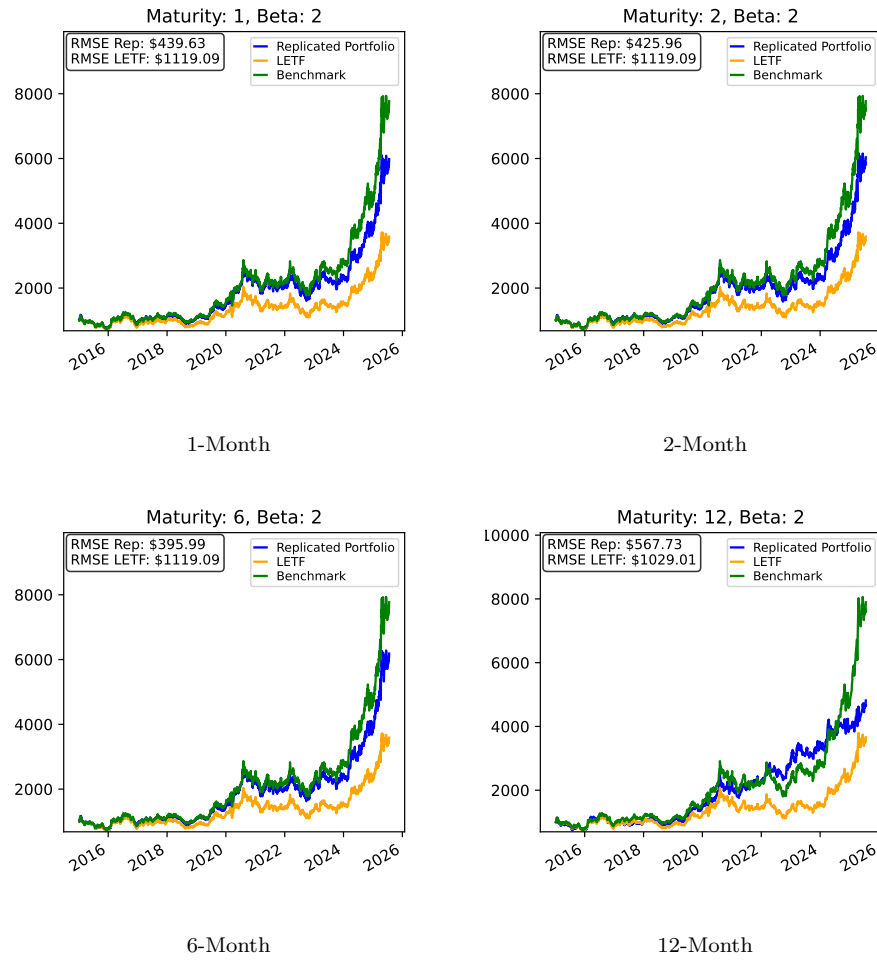


Figure 7: **UGL (2x) Replication:** The replicated portfolios exhibit significantly greater performance than the leveraged ETF. Once again, the 12m maturity performs worse than all others.

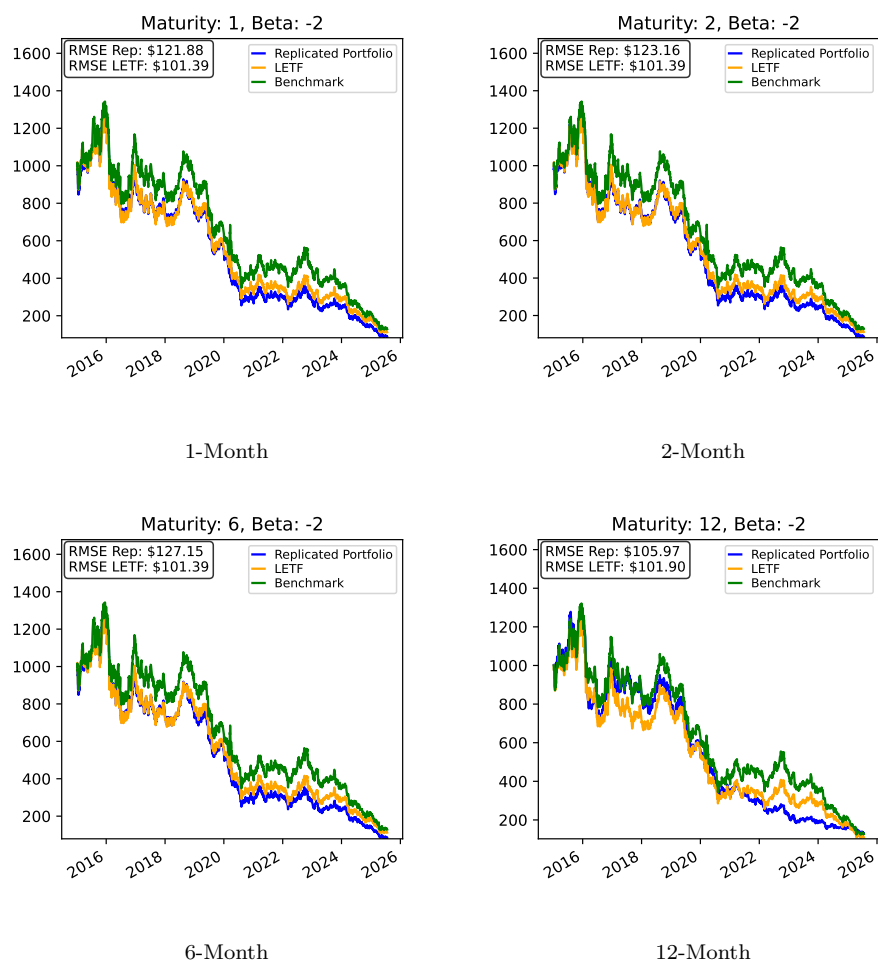


Figure 8: **GLL (-2x) Replication**: Better performance for all replicated portfolios compared to the leveraged ETF, but the 12m maturity seems less reactive to spot.

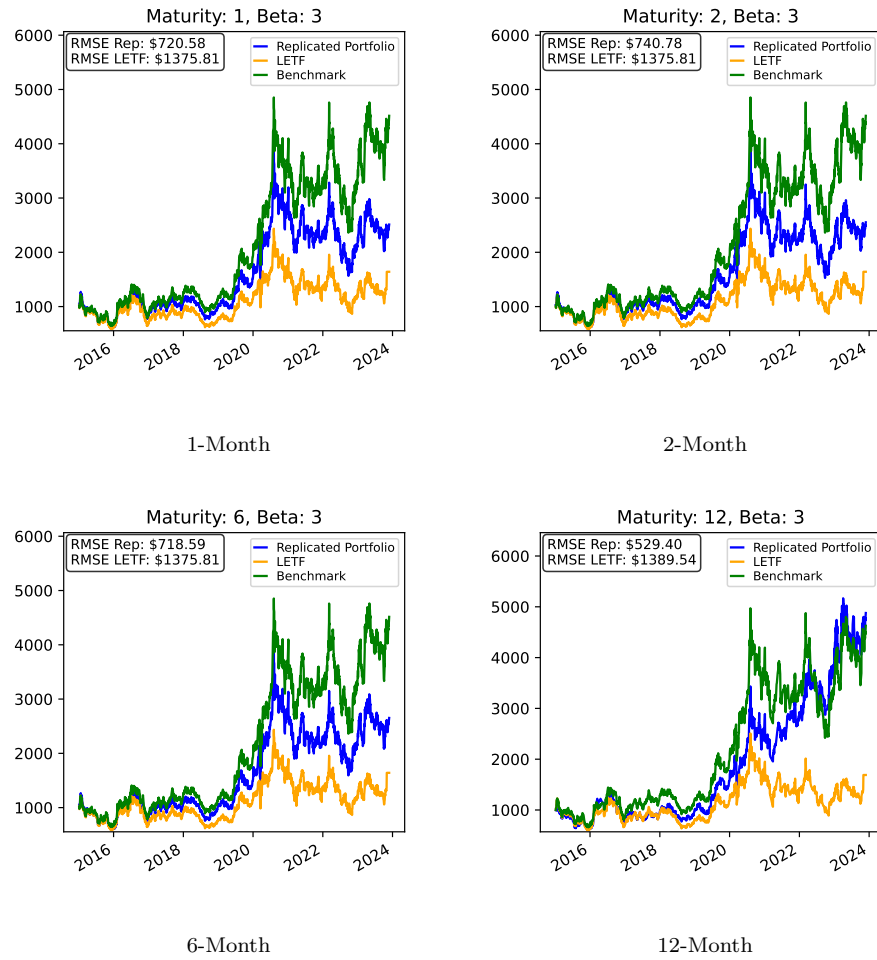


Figure 9: **UGLDF (3x) Replication**: All replicated portfolios performed better than the LETF. The higher leverage (3x) amplifies the divergence errors, particularly in the longer-dated contracts.